

Contents

Command Reference — CCNA Labs (01-06)	1
1. Moving between IOS modes	1
2. Basic device setup	1
3. Interface configuration	1
4. Switching (VLANs & trunks)	2
5. Routing	2
6. DHCP (router as server)	2
7. NAT / PAT	2
8. Verification / show commands	2
9. Removing / undoing config	3
10. PC commands (Desktop → Command Prompt)	3
11. Mac Terminal (shell) commands	3
12. Git commands	3

Command Reference — CCNA Labs (01-06)

Every command used across the labs so far, grouped by purpose. Cisco IOS commands run on routers/switches; PC commands run in the Desktop → Command Prompt; shell/git commands run in the Mac Terminal.

1. Moving between IOS modes

Command	What it does
enable (en)	Enter privileged EXEC mode (R1#)
configure terminal (conf t)	Enter global config mode (R1(config)#)
interface gigabitEthernet0/0	Enter interface config (R1(config-if)#)
interface range fastEthernet0/1 - 2	Configure several ports at once
interface gigabitEthernet0/0.10	Create/enter a subinterface
interface loopback0	Create/enter a virtual loopback
router ospf 1	Enter OSPF router config (R1(config-router)#)
ip dhcp pool LAN-P00L	Enter DHCP pool config
exit	Go back one mode level
end	Jump straight back to privileged EXEC

2. Basic device setup

Command	What it does
hostname R1	Name the device
copy running-config startup-config (wr)	Save config so it survives reload

3. Interface configuration

Command	What it does
ip address 192.168.1.1 255.255.255.0	Assign an IP + mask
no ip address	Remove the IP from an interface
no shutdown	Turn the interface on
shutdown	Turn the interface off
description LINK-T0-LAN	Add a label (cosmetic)

Command	What it does
ip nat inside / ip nat outside encapsulation dot1q 10	Mark the private / public NAT side Tag a subinterface for a VLAN

4. Switching (VLANs & trunks)

Command	What it does
vlan 10	Create VLAN 10
name IT	Name the VLAN
switchport mode access	Set port as an access (end-device) port
switchport access vlan 10	Put the port in VLAN 10
switchport mode trunk	Set port as a trunk (carries all VLANs)
switchport nonegotiate	Disable DTP auto-negotiation

5. Routing

Command	What it does
ip route 192.168.2.0 255.255.255.0 10.0.0.2	Static route to a network
ip route 0.0.0.0 0.0.0.0 200.0.0.2	Default route (gateway of last resort)
router ospf 1	Start OSPF process 1
network 192.168.1.0 0.0.0.255 area 0	Advertise a network in OSPF area 0

6. DHCP (router as server)

Command	What it does
ip dhcp excluded-address 192.168.10.1 192.168.10.10	Reserve IPs DHCP won't hand out
ip dhcp pool LAN-P00L network 192.168.10.0 255.255.255.0	Create an address pool Range of addresses to lease
default-router 192.168.10.1 dns-server 8.8.8.8	Gateway given to clients DNS server given to clients

7. NAT / PAT

Command	What it does
access-list 1 permit 192.168.1.0 0.0.0.255	Define which addresses may be translated
ip nat inside source list 1 interface gigabitEthernet0/1 overload	Translate (PAT) matched traffic to the outside interface IP

8. Verification / show commands

Command	Shows
show ip interface brief	Every interface: IP + up/down status
show ip route	The routing table (C, S, O, L codes)

Command	Shows
show ip protocols	Which routing protocols are running
show ip ospf neighbor	OSPF neighbors and their state (FULL)
show vlan brief	VLANs and which ports are in each
show interfaces status	Per-port status, VLAN, connected/notconnect
show interfaces trunk	Which ports are trunks and allowed VLANs
show interfaces gig0/1 switchport	Detailed switchport mode info
show cdp neighbors	Directly connected Cisco devices + ports
show ip nat translations	Active NAT translations
show ip nat statistics	NAT inside/outside interfaces + hit counts
show access-lists	Configured ACLs and their entries
show running-config	The full active config
show running-config \> include nat	Filter the config for matching lines

9. Removing / undoing config

Put no in front of most commands to reverse them:

Command	What it does
no shutdown	Enable an interface (undo shutdown)
no ip address	Remove an IP
no ip route 192.168.2.0 255.255.255.0 10.0.0.2	Delete a static route
no network 10.0.0.2 0.0.0.3 area 0	Remove an OSPF network line

10. PC commands (Desktop → Command Prompt)

Command	What it does
ipconfig	Show the PC's IP, mask, gateway
ping 192.168.2.2	Send 4 test packets to a host
ping -t 8.8.8.8	Ping continuously (Ctrl+C to stop)
ping -n 10 <ip>	Send a specific number of packets

11. Mac Terminal (shell) commands

Command	What it does
cd <folder>	Change directory
ls / ls -a	List files (-a includes hidden ones)
pwd	Print the current folder path
mkdir <name>	Make a folder
touch <file>	Create an empty file
mv <src> <dest>	Move / rename a file
cp <src> <dest>	Copy a file
cat <file>	Print a file's contents
head -3 <file>	First 3 lines of a file
grep "text" <file>	Search a file for text (-A5 = 5 lines after)
find ~ -name "file" 2>/dev/null	Search for a file by name
nano <file> / open -e <file>	Edit a file (nano / TextEdit)

12. Git commands

Command	What it does
<code>git init</code>	Start tracking a folder
<code>git add .</code>	Stage all changes
<code>git commit -m "message"</code>	Save a snapshot with a message
<code>git push</code>	Upload commits to GitHub
<code>git pull --no-edit</code>	Download + merge remote changes
<code>git config pull.rebase false</code>	Set merge as the pull strategy
<code>git remote add origin <url></code>	Link the repo to GitHub
<code>git remote -v</code>	Show the linked remote
<code>git remote set-url origin <url></code>	Change the remote URL
<code>git status</code>	Show what's changed / staged
<code>git log --oneline</code>	Compact commit history
<code>git ls-files</code>	List tracked files
<code>git rm --cached <file></code>	Stop tracking a file (keep it on disk)
<code>git branch -M main</code>	Rename the current branch to main
<code>git mv <old> <new></code>	Rename/move a tracked file